

JIMI ADEMOLA

Canada

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Education

McGill University

Masters of Science in Computer Science

Sep. 2026 - Ongoing

Montréal, QC

University of British Columbia

Bachelor of Science in Computer Science (Honours)

Sep. 2022 – May 2026

Kelowna, BC

Experience

University of British Columbia

Research Assistant

Sep 2025 – Ongoing

Kelowna, BC

- Engineered automated static analysis pipelines using SonarQube to extract code quality metrics from over 1,900 merged Pull Requests.
- Modeled PR integration probabilities by evaluating repository metadata with Generalized Linear Mixed Models (GLMM) in R.
- Evaluated differences in code quality metrics across mature repositories using nonparametric statistical tests.

Natural Sciences and Engineering Research Council of Canada

Undergraduate Developer / Researcher

May 2025 – Aug. 2025

Kelowna, BC

- Contributed to research on higher-order generators with implications on cryptography and encryption.
- Utilized and built CRAN packages in order to discern variables in industry-standard random number generators.
- Created 5+ algorithms to attack statistical weaknesses in Wichmann-Hill, Mersenne-Twister and other Combined Congruential Generators.
- Assisted research group in novel wildfire modeling techniques including data scraping, analysis and simulation using R and Python.

Projects

LearnCoding | PHP, Laravel, MySQL

- Led deployment of the LearnCoding platform to public environments, ensuring smooth CI/CD integration and uptime.
- Implemented cybersecurity scanning tools and protocols to identify vulnerabilities and safeguard user data.
- Developed backend services using PHP and the Laravel framework to support authentication, course delivery, and user management.

Project Artifact Miner | Python, Electron, FastAPI

- Developed a static software miner utilizing machine learning models to analyze a user's projects and generate insights
- Produced both a *pip* package for command line use and a front-end powered by Electron
- Implemented parallel computing principles to produce a 300% speedup in file processing time
- Generated 20+ statistics and insights for display on a user's portfolio or resume

GoGuard IPS | Go, Kali Linux, Ubuntu

- Engineered a real-time Intrusion Prevention System in Go with a five-stage packet inspection pipeline and automated `iptables` enforcement
- Developed stateful TCP flow reassembly to detect fragmentation-based DPI evasion across split payloads spanning multiple packets
- Implemented MAC-IP binding spoofing detection cross-referenced against the kernel ARP table for attribution

Research Contributions

Beyond the Pull Request: Examining Code Quality in Mature Open Source Projects | (2026) In preparation

[In]Secure: Using Statistical Weaknesses to Crack Pseudo-Random Number Generators | (2025) Unpublished

Technical Skills

Languages: Python, Java, C, HTML/CSS/JavaScript, TypeScript, SQL, R, PHP, React

Developer Tools: VS Code, Docker, Google Cloud Platform, Android Studio, Git/GitHub, Jupyter Notebook

Technologies/Frameworks: Linux, Jenkins, JUnit, WordPress, MITRE ATT&CK